🔑 **Answer Key — Teacher Copy · Lesson 6-15**

MSTAR PRACTICE WORKSHEET

Lesson 6-15: Simplify Algebraic Expressions

Grade 6 Mathematics · Standard 6.AT.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Simplify 7x + 3x.

- **A** 10x
- **B** $10x^2$
- **C** 21x
- **D** 73x

7x + 3x = 10x. Add the coefficients: 7 + 3 = 10.

Question 2 SELECTED RESPONSE — CORRECT: A

Simplify 9n + 4 + 2n.

- **A** 11n + 4
- **B** $11n^2 + 4$
- **C** 15n
- **D** 92n + 4

9n + 2n = 11n. The constant 4 has no like term, so it stays. Result: 11n + 4.

Question 3 SELECTED RESPONSE — CORRECT: A

Which pair are like terms?

- (A) 6x and 2x
- (B) 6x and 6y
- (C) 6x and 6
- (D) 6x and $2x^2$

6x and 2x are like terms because they both have the same variable (x) raised to the same power (1).

Question 4 SELECTED RESPONSE — CORRECT: A

Simplify 5a + 3 + 2a.

- (A) 7a + 3
- (B) $7a^2 + 3$
- (C) 10a
- (D) 52a + 3

5a + 2a = 7a. The constant 3 stays. Result: 7a + 3.

Question 5 SELECTED RESPONSE — CORRECT: A

The board shows 3x + 5x on one channel. Simplify it.

- (A) 8x
- (B) $8x^2$
- (C) 15x
- (D) 35x

3x and 5x are like terms, so add the coefficients: 3 + 5 = 8. The variable x stays, giving 8x.

Question 6 SELECTED RESPONSE — CORRECT: A

Simplify the track: 4a + 2 + a.

A 5a + 2

B $5a^2 + 2$

C 6a

D 7a

4a + a = 5a (remember a means 1a). The 2 has no like term, so it stays. Result: 5a + 2.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Simplify the expression 8x + 3 + 2x – 1.

A. 6x + 2

B. 10x + 2

C. 10x + 4

D. 12x

The terms 8x and 2x are like terms, so add their coefficients: 8 + 2 = 10, giving 10x. The constants 3 and –1 combine to 2. The simplified expression is 10x + 2.

- **A:** The x terms were subtracted. Both are positive, so 8 plus 2 gives 10x.
- **C:** The constants were added as 3 plus 1. The second is minus 1, so they combine to 2.
- **D:** The constants were folded in with the x terms. 3 and negative 1 are not like terms with 8x.

Part B — correct answer: A

Which reasoning shows why your answer to Part A is correct?

A. 8x and 2x share the same variable part, so add their coefficients to get 10x. The constants 3 and –1 combine separately to 2, giving 10x + 2.

B. Combine every number in the expression into one coefficient: 8 + 3 + 2 – 1 = 12, giving 12x.

C. Subtract the coefficients of the variable terms: 8 – 2 = 6, giving 6x + 2.

D. Add the coefficients of the variable terms and then add both constants: 10x + 4.

Only terms with the same variable part can be combined, and constants combine with constants. The constant 1 is being subtracted, so 3 – 1 = 2, and the x terms are added, not subtracted.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the expression 5y + 7 + 3y + y.

- **A** 5y, 3y, and y are like terms.
- **B** The coefficient of the term y is 1.
- **C** The simplified expression is 9y + 7.
- **D** The simplified expression is 16y.
- **E** 7 is a constant and cannot be combined with the y terms.
- **F** The simplified expression is 8y + 7.

All three variable terms have the same variable part y, so they are like terms. A term written as y has an understood coefficient of 1, so 5 + 3 + 1 = 9, giving 9y. The constant 7 has no variable and stays separate, so the simplified expression is 9y + 7. The answer 8y + 7 leaves out the coefficient 1, and 16y wrongly folds the constant into the variable terms.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Three students simplified 4x + 3 + 2x + 5 + x differently. Student A got 7x + 8. Student B got $7x^2 + 8$. Student C got 6x + 9. Who is correct? Explain the mistakes the other two students made.

Model answer: Student A is correct: 4x + 2x + x = 7x (the lone x counts as 1x) and 3 + 5 = 8, giving 7x + 8. Student B added the coefficients correctly but changed x into x^2, which combining like terms never does, and Student C left out the lone x and instead pulled a 1 into the constants, landing on 6x + 9.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.